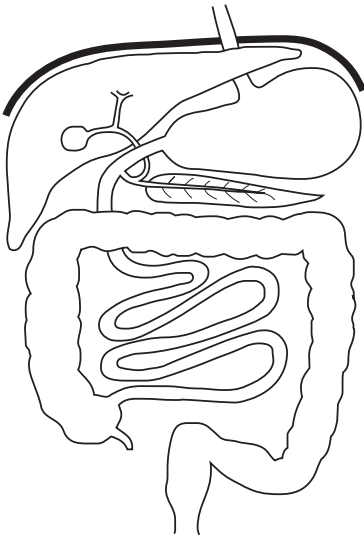


6. The diagram below shows part of the human digestive system.

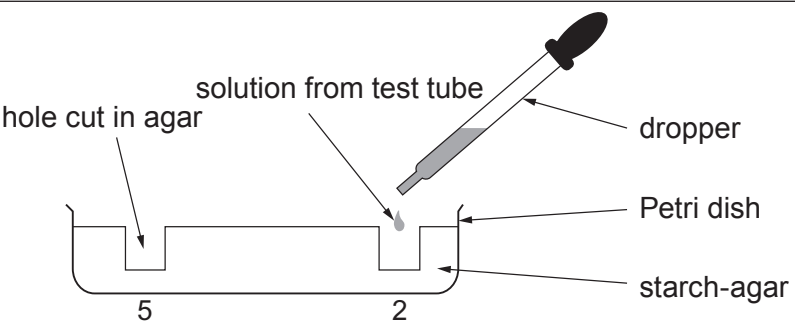


- (a) On the diagram above label the following: [2]
- A** pancreas
B bile duct
- (b) The diagram below shows five test tubes **A – E** and their contents. The pH and temperature of each is also shown.

	Solution				
	Distilled water	0.5% amylase	1% amylase	1% amylase	2% amylase
tube					
temperature (°C)	37	37	100	37	37
pH	7	7	7	7	7

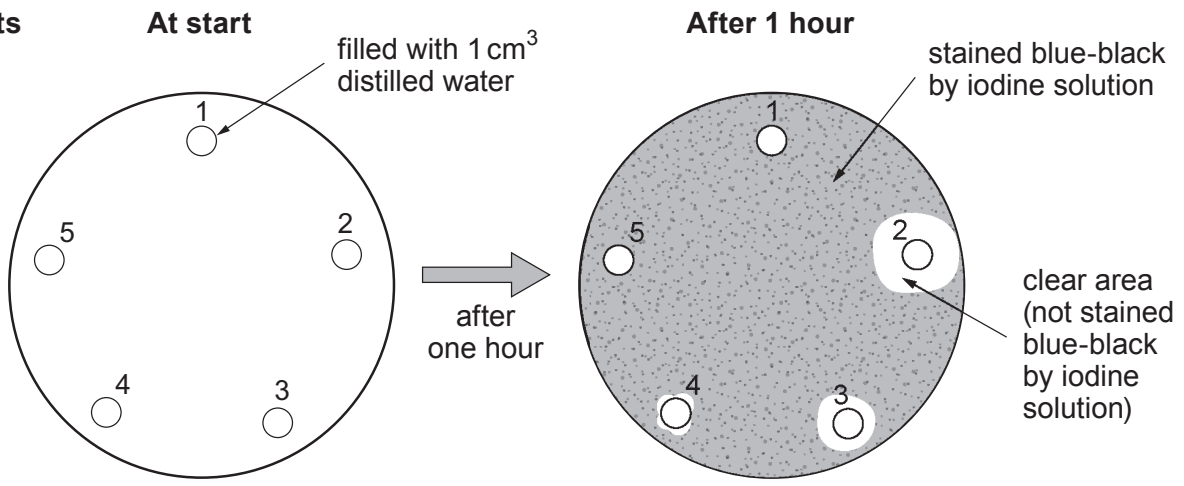
- A pupil carried out the following procedure:
- She took a Petri dish containing starch-agar (a set jelly in which starch had been dissolved).
 - She cut five small holes in the jelly (holes 1 – 5).
 - She put 1 cm³ distilled water from Tube **A** into hole 1.
 - Holes 2 – 5 were each filled with 1 cm³ of a different solution from one of test tubes **B**, **C**, **D** or **E**.
 - After one hour she flooded the Petri dish with iodine solution.





Cross-sectional diagram showing two of the five holes.

Results



(i) Explain the appearance of the clear areas. [1]

.....

.....

(ii) State the solution that was added to hole 2. Explain your answer. [2]

.....

.....

(iii) Which hole contained the contents of tube C? Explain your answer. [4]

Hole number

Explanation

.....

.....

.....

(c) State the purpose of tube A, the contents of which was put into hole 1. [1]

.....

10

